

Deep Learning in Recommender Systems

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Recommender systems have undertaken significant advances over the last years, evolving from collaborative filtering techniques to deep learning architectures capable of modeling complex and multimodal interactions. The increasing adoption of graph neural networks (GNNs), recurrent models, attention mechanisms, and, more recently, large language models (LLMs), has notably strengthened the capacity of state-of-the-art recommender systems to model user behavior and deliver more relevant and personalized recommendations. This Special Issue on Deep Learning in Recommender Systems presents a series of contributions that evidence this transformation, addressing methodological innovation, domain-specific deployment, and emerging concerns such as fairness and responsible recommendation.

1. Graph-Based Deep Learning

Graph Neural Networks (GNNs) have become a powerful approach in artificial intelligence for modeling structured data organized in graph form, enabling effective modeling of complex structured relationships. In recommender systems, they use user–item interaction graphs to learn rich representations that improve link prediction and enable more accurate, personalized recommendations. A significant limitation of GNNs is their susceptibility to biases present in the training data. Addressing this issue has therefore become a critical focus for researchers in the field. Some of the contributions in this Special Issue reflect the consolidation of this paradigm and propose novel strategies to address key challenges associated with GNNs.

Li et al. [1] introduce MIMA, a multi-feature interactive meta-path aggregation algorithm based on heterogeneous graph neural networks, designed to effectively capture semantic relationships in heterogeneous information networks, such as user–item graphs enriched with multiple attributes. By aggregating semantic information derived from meta-paths, the proposed model enhances interpretability while preserving structural and attribute-level information. Experimental results demonstrate that fully leveraging user and item features leads to significant improvements in recommendation accuracy.

Graph convolutional networks have also been recently applied to collaborative filtering. However, this approach suffers from over-smoothing, which makes node representations overly similar and reduces the ability to distinguish between users and items. Ou and Niu [2] address this drawback and structural limitations in graph collaborative filtering by proposing the dual-channel feature-enhanced collaborative filtering (DCFECF) model. The local convolutional channel incorporates a self-attention mechanism, while the global channel applies K-means++ and fuzzy C-means clustering to build a more complete interaction graph and avoid local optima. Experiments demonstrate that DCFECF effectively improves the performance of the original DCCF model.



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Temporal and session-aware dynamics are addressed by Lin et al. [3], who investigate GRU-based latent interaction integration for session-aware recommendation as an alternative to session-based recommendations. The proposal integrates latent-context features with both long-term and short-term user preferences. By enhancing contextual modeling through Gated Recurrent Units (GRU) and attention mechanisms, the work demonstrates the importance of short-term behavioral dynamics in optimizing recommendation performance.

Beyond structural and temporal improvements, fairness and bias mitigation emerge as critical dimensions in graph-based recommender systems. Chizari et al. [4] examine inter-bias effects and fairness–accuracy trade-offs within GNN-based recommendation models. Their analysis highlights how different bias types can enhance or mask each other, as well as affect predictive performance. The proposal includes a method for evaluating the recommendation model performance across diverse demographic groups while controlling for the confounding influence of distinct sources of bias. This contribution reflects an important shift in the field, moving away from accuracy-driven optimization toward multi-objective evaluation that also considers ethical aspects.

These works illustrate the evolution of graph-based deep learning approaches: from modeling increasingly complex relational patterns to integrating fairness-aware analysis within the evaluation process.

2. Language Models and Hybrid Representation Learning

Another major research direction captured in this Special Issue is the integration of language models and semantic representations into recommender systems.

Shehmir and Kashef [5], in their comprehensive survey LLM4Rec, systematically analyze the incorporation of large language models into recommendation pipelines, specifically in content-based approaches. Their work synthesizes architectural strategies, application domains, and open challenges. The survey underscores the potential of LLMs to enrich user inputs, leverage heterogeneous and unstructured data, and mitigate classical recommender system challenges, among other capabilities. This, in turn, enables reasoning over textual content and supports conversational and explainable recommendation scenarios.

Complementing this perspective, Li et al. [6] evaluate several types of embeddings derived from pre-trained language models and knowledge graphs for educational content recommendation, with the aim of identifying which representations are most suitable for the task. In addition to individual models, they also assess ensemble approaches based on different combination strategies. Their empirical comparison highlights the value of hybrid semantic representations in improving content linkage and recommendation quality. Overall, this work provides concrete evidence of how transfer learning and pre-trained models can enhance domain-specific recommendation tasks.

Together, these contributions suggest that recommender systems are entering a new phase in which semantic understanding, cross-modal integration, and foundation models will play a central role in representation learning and inference.

3. AI-Driven Recommendation in Education

Education appears as a notable application domain in this Special Issue, with potential social impact. Dahal et al. [7] present an AI-supported learning recommendation framework integrated into Moodle and extended toward Learning eXperience Platforms (LXPs). Their work demonstrates how deep learning techniques can be operationalized in real higher education environments to personalize learning pathways and improve student engagement.

In addition to addressing LLMs, as discussed in the previous section, Li et al. [6] also explore educational recommendations, focusing on linking and recommending learn-

ing materials. These contributions also consider education as a context for responsible and explainable recommender systems. In such settings, recommendation outcomes can influence learning paths, making considerations like fairness, interpretability, and robustness relevant.

4. Emerging Trends

The articles collected in this Special Issue collectively reveal several overarching trends:

- Consolidation of graph neural architectures for modeling complex and heterogeneous interactions [1–4].
- Integration of foundation models and semantic embeddings into recommendation pipelines [5,6].
- Growing emphasis on fairness-aware and multi-objective evaluation, particularly within deep graph-based systems [4].
- Expansion of high-impact application domains, notably education [6,7].

Based on the trends identified above, it can be concluded that current research in recommender systems is increasingly oriented toward combining advanced graph-based architectures with foundation models and semantic embeddings, while increasingly considering fairness and multi-objective evaluation. Applications are expanding into socially impactful domains like education, highlighting the need for responsible, transparent, and explainable systems. Future work is likely to focus on explainability, richer evaluation protocols, multimodal and cross-domain approaches, and the efficient integration of large foundation models, with responsible AI principles embedded throughout.

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